

**Enhancing MobileNetV3-Small Architecture Through Integration of D-CLAHE and
Layerwise Progressive Fine Tuning for Image Classification**

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Convolutional Neural Networks (CNN) are a major breakthrough of human technology in the field of Artificial Intelligence (AI) and Deep Learning, this enables computers to mimic human vision analyzing visual data like images and videos. This features classification tasks, automating manual labors like visual inspections of multiple parameters and features (What Is a Convolution Neural Network? | Glossary, n.d.). In recent years, these CNNs are now being put to our mobile devices and are now capable of running them, one of these CNN models is the MobileNet version 3 small variant.

The MobileNetV3 Small version, with 2.54 million parameters it is efficient and fast with inference times as low as 0.536 ms on certain devices and is being used even with the release of V4 model (MobileNet V3 Small · Models · Dataloop, n.d.). Compared to other image classification models, MobileNetV3 small require lower parameter count and lower computational power unlike ResNet-50 and VGG 60 making the other models incompatible as image classification model to limited processing power like mobile phones (Ren et al., 2025). Yet with all of that, MobileNetV3 Small still needs improvement in real world application for better edge detection utilizing histogram equalization (Leveraging MobileNetV3 for In-Field Tomato Disease Detection in Malawi via CNN, n.d.) and the use of extra fine tuning for better model performance (Zhang et al., 2025). Restrict fine-tuning to upper layers to avoid catastrophic forgetting of pretrained representations (Domain-Adapted MobileNetV2 & V3, n.d.)

Contrast Limited Adaptive Histogram Equalization or CLAHE is a type of image processing technique used to improve contrast and enhance edges without amplifying noise. However, conventional CLAHE typically relies on a fixed global clip limit regardless of the histogram distribution of each local region, which frequently leads to unnatural over-enhancement in already bright areas, thus Differentiable CLAHE was introduced. Differentiable CLAHE address this limitation by adaptively estimating tile-wise clip limits directly from the input image . Instead of relying on static parameters, recent frameworks uses lightweight Convolutional Neural Networks (CNNs) to calculate these local adjustments dynamically. This enhancement mechanism can be completely integrated into end-to-end network optimization making the preprocessing an active, learnable layer within the neural network. Because the module can participate in forward and backward propagation during each training epoch, the network learns to continuously optimize its own brightness and contrast parameters based on the specific loss calculations (Otsuka et al., 2026) . On the other side, the Layer Wise Progressive Fine Tuning will be implemented to reduce layer freezing or the forgetting of the MobileNetV3 small of the original trainings when new data is trained per layer (Efficient Layer-wise LLM Fine-tuning for Revision Intention Prediction, n.d.). Using the Layer Wise Progressive Fine Tuning, the layers will be unfrozen layer by layer starting from the high level semantic classifiers moving down until the feature extractors, this allows the model to retain its generalized pre trained edge detection weights on previous trainings while slowly adapting to the new data (Zaoyang, 2025). Paired with the ADAM optimizer, this layers unfreezing prevents gradient shock, significantly reducing validation loss and ensuring that the MobileNetV3-Small architecture achieves

generalization without requiring a massive, local dataset (Improved Adam Optimizer for Deep Neural Networks, n.d.).

This study proposed an enhancement to the architecture for the MobileNetV3 Small by integrating the Differentiable Contrast Limited Adaptive Histogram Equalization that will be created and implementing Layerwise progressive Fine tuning to prevent gradient shock and stabilize the network. The D-CLAHE module will be integrated in the first layer that utilizes an internal convolutional sub-network to dynamically learn and optimize brightness and contrast parameters for individual input images, adjusting autonomously based on the network's classification loss during each training epoch that will be included during the forward and backward propagation.

1.2 Research Objectives

General Objectives

The study aims to develop an enhanced MobileNetV3-Small using a Differentiable Contrast Limited Adaptive Histogram Equalization (D-CLAHE) and Progressive Fine-Tuning to improve the model's performance.

Specifically, this study aimed to:

Specific Objectives

1. Develop an enhanced MobileNet v3 Small model integrating D-CLAHE and Progressive Fine Tuning.
2. Compare the performance of the base MobileNet v3 Small to the enhanced model in terms of the following measures:

2.1 Training Time

2.1.1 Training Time

2.1.2 Memory Usage

2.1.3 Weight Discrepancy (L2 Norm)

2.2 Classification model evaluation metrics

2.2.1 Accuracy

2.2.2 Precision

2.2.3 Recall

2.2.4 F1 Score

2.2.5 True Negative Rate

2.2.6 Macro F1 score

2.2.7 Matthews Correlation Coefficient

2.3 Inference Performance Metrics

2.3.1 Inference Latency

2.3.2 Multiply-Accumulate Operations

2.3.3 Model Size

3. Verify the significance level using significance tests

4.1 Welch T test

4. Deploy the enhanced model to an android mobile application for compatibility

1.3 Significance of the Study

The study focuses on the improvement of the MobileNetV3 Small image classification algorithm by integrating D-CLAHE to address its weakness on natural

lights combined with progressive fine tuning. With this, the methodology and results will be beneficial to the following sectors:

1. Computer Science and IT community - The proposed study addresses the limitations of the MobileNetV3 Small Model when applied in the real world tests using D-CLAHE can help the CS and IT community to explore other models vulnerabilities and find possible solutions to fix the issues and can be started using the proposed methodologies.
2. Farmers - The proposed deployed application with the enhanced model can help the farmers to identify the cassava leaf diseases to increase the possible revenue of the farmers and reduce the wasted crops.
3. Future Researchers - The literatures provided in this study can help other researchers to establish a literature review for their related studies. This in turn can help them to establish the importance of the model or utilization of the proposed methodology in past papers.
4. College of Computing Studies and Information Communication Technology - The study can help the department to inspire the students to appreciate the process of model enhancements and the process needed to achieve it.

1.4 Scope and Limitations

This study focused on enhancing MobileNetV3-Small by integrating a Differentiable Contrast Limited Adaptive Histogram Equalization (D-CLAHE) to the

layer 0 of the neural network and Layerwise Progressive Fine Tuning to improve real-world performance of the model when applied to a mobile device.

This study uses the different evaluation metrics to check the improvements of the proposed enhanced model to the base model, training time to check if there is a difference in time needed for training, memory usage to check the amount of memory needed in training and Weight discrepancy in L2 norm to validate the retention of weights in the fine tuning. As for the evaluation, the study uses accuracy to compare the overall classification of both model, precision to check how exact and consistent the models in correct classifications, recall to evaluate both model's ability to identify all relevant instances of a class, f1 score to validate the results in the possible data imbalances, MacroF1 to evaluate the minority classes in the results of both models and MCC score to compare the overall performances of both models into a single metric. The study will also use the inference performance metrics to check the applied models performance in the deployed mobile application, Inference latency to check if theres a difference in the classification delay, MAO for CPU usage comparison and the resulting model size.

The study will use a significance test to validate the impact of the enhancements to the base model, Welch's t-test is applied to the accuracy distributions of multiple training runs and a p value of less than 0.05. -

The study will use the Cassava Leaf Diseases; Cassava Mosaic Disease (CMD), Cassava Bacterial Blight (CBB), Cassava Witches' Broom Disease (CWBD), Brown Leaf Spot (*Cercospora henningsii*), and Healthy (Asymptomatic) leaves as a dataset for training, evaluation, model testing and field testings

The study did not extend to implementing other image enhancement in preprocessing and will only have the basic image reduction and will rely on the added layer of D-CLAHE. Also the study will only use the layer wise progressive fine tuning paired with an adam optimizer to balance the weights.

The enhancement, training, fine-tuning, and evaluation of the enhanced MobileNetV3-Small model and base model were done on specific hardware. This will affect the measurements of inference latency and training times therefore, this study could not provide measurements directly comparable to those of experiments and studies that do not utilize the hardware described in this study.

The study could not make recommendations on the offline mobile application's minimum or recommended system specifications, as this would require further testing across different mobile hardware platforms. Furthermore, the application's consumption towards components other than inference latency and storage size was not measured in the study, making it impossible to draw conclusions that require those measurements.

CHAPTER 2

THEORETICAL FRAMEWORK

2.1 Review of Related Literatures

2.1.1 Convolutional Neural Network

Convolutional Neural Networks (CNNs) are deep learning models that interpret visual data by automatically learning spatial feature hierarchies across multiple layers. These models have acquired favor in agriculture, particularly in diagnosing diseases, where traditional methods fail because of the similarity of visual symptoms between different diseases. Unlike traditional artificial neural networks, CNNs preserve the spatial hierarchy of pixels by passing visual filters over the input image (H.ning, et al, 2023).

In a standard classification architecture, after the CNN backbone extracts the spatial features, the data enters a Dense Fully Connected layer. The final decision is computed using the Softmax activation function, which converts the raw output logits into a normalized probability distribution that sums to 1. To optimize the network during training, the Categorical Cross-Entropy loss function is utilized to measure the divergence between the predicted probability distribution and the actual true labels as ground truth, driving the Adam optimizer to update the network weights and minimize classification errors over successive epochs (Li et al., 2025; Abiyyu & Rahardi, 2026; Sanida & Dasygenis, 2024).

Optimization algorithms are deployed to minimize this error via backpropagation. The study highly recommends Adam (Adaptive Moment Estimation) optimizer over

traditional Stochastic Gradient Descent (SGD) for image classification. Adam computes individual, adaptive learning rates for different network parameters based on estimates of first and second gradients. This adaptive scaling prevents the model from getting stuck in local minima and is particularly crucial when fine-tuning lightweight architectures like MobileNetV3-Small on highly localized datasets (Srinivas et al., 2024; Xu et al., 2023).

2.1.2 MobileNetV3 Small

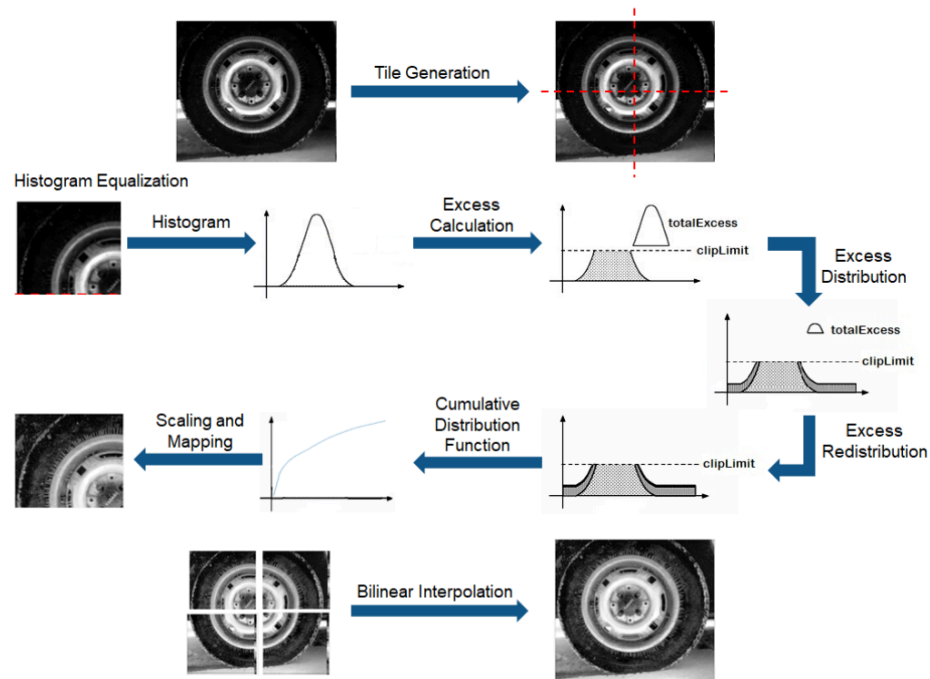
Under MobileNet is its third generation the V3, with the release of V3 comes the small and large and can be used in different cases MobileNetV3-Large is optimized for higher performance and is suitable for devices with more computational resources. On the other hand, MobileNetV3-Small is designed for low resource devices, offering a more compact model with reduced computational complexity. Both variants provide state-of-the-art results in mobile classification, detection, and segmentation tasks, making them suitable for a wide range of applications (Activeloop, n.d.).

An improvement in MobileNetV3-Small is the replacement of computationally expensive nonlinearities like standard Swish or Sigmoid with modified, hardware-friendly versions. Using the hswish on the second half of the convolution network which is a piece-wise linear approximation mobilenet v3 small can increase accuracy with no latency increase, showing a 2.5 million parameters with 57 million multiply adds.

2.1.3 Contrast Limited Adaptive Histogram Equalization

Figure 2

CLAHE Architecture



CLAHE is a preprocessing step used to improve image quality before training models, resulting in higher accuracy, better generalization, and enhanced feature extraction. The process consists of 3 parts, the tile generation, histogram equalization and the bilinear interpolation. As shown in figure 2, the image is first separated into 4 tiles, Histogram equalization is performed on each tile using the define clip limit. The histogram equalization which is the main feature of CLAHE has five parts. The first process on the 1st tile is the histogram computation, this is the analysis of the tile itself on the contrasts of the bright and dark spots, second is the excess calculation, on this part the histogram removes the total excess based on the defined clip limit then an excess distribution to the clip limit to equalize the bright within limit and darks within limits on the excess

distribution, then lastly the cumulative distribution ensured the equal distribution curve on the image for scaling. After the histogram evaluation on every tile, the resulting tiles are stitched together using bilinear interpolation then a final image check for equal gradient of the image contrasts (*Contrast Limited Adaptive Histogram Equalization - MATLAB & Simulink*, n.d.), particularly in CT scans, MRIs, and X-ray imaging. Since the application of CLAHE is tested on these field, it is the next step to find other areas like in the agriculture, with the flexibility of the model on its parameters such as the clip limit and grid size, it can be used in various environment or light conditions where the agricultural sector experiences (Okyere-Gyamfi et al., 2025).

In this research, the authors constructed a custom architecture heavily inspired by the MobileNetV3-Small backbone, utilizing Inverted Residual blocks, depthwise separable convolutions, and SE modules to keep the final model size under 2 Megabytes. While this ultra lightweight footprint was perfect for smartphone deployment, the researchers noted that the network's restricted depth made it highly susceptible to a texture bias, meaning the model struggled to differentiate between the subtle color gradients of early stage fungal infections and natural leaf discoloration caused by sun glare. The dataset was then processed using CLAHE and the researchers detailed how CLAHE's localized tile-processing and strict "clip limit" prevented the over-amplification of homogeneous green leaf areas, while dramatically sharpening the edges and enhancing the localized contrast of the disease lesions. Because MobileNetV3-Small relies on a h-swish activation function and highly compressed feature maps to save processing time, it requires highly distinct input manifolds to trigger strong gradient updates during

training. The application of CLAHE provided exactly this, it separated the foreground disease features from the background leaf tissue (MDPI, 2025).

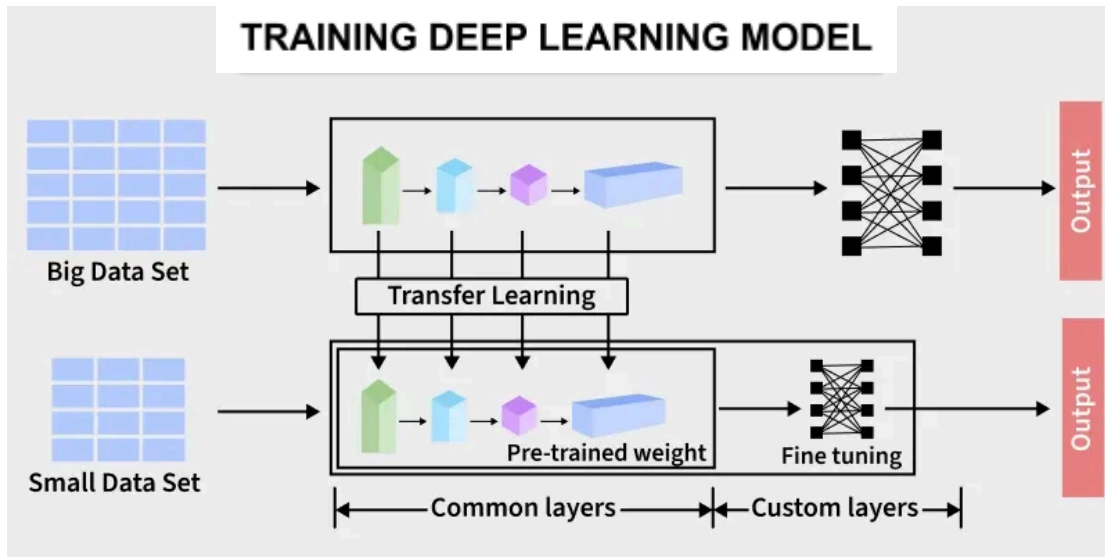
2.1.4 Progressive Fine Tuning

Progressive Fine tuning in image processing is a transfer learning technique where a pre-trained model is adapted to a new, specialized task in multiple, staged steps rather than all at once. This method allows the model to learn complex or domain-specific features gradually, starting with broader, lower-level features before focusing on fine-grained, high-level details. The primary function of this is to help the model not forget the previous training when new data was used for much more complex operation, this situation is called catastrophic forgetting. It is highly effective for reducing over-fitting, optimizing resource allocation that is typically encountered in highly changing or large datasets (Ji et al., 2025).

Fine-tuning allows a pre-trained model to adapt to a new task. This approach uses the knowledge gained from training a model on a large dataset and applying it to a smaller, domain-specific dataset. Fine-tuning involves adjusting the weights of the model's layers or updating certain parts of the model to improve its performance on the new task (GeeksforGeeks, 2025).

Figure 3

Fine tuning model architecture



The progressive fine tuning architecture is especially crucial for lightweight models like MobileNetV3-Small. The improvement has demonstrated that lightweight models, which use very few parameters, easily overfit and suffer from "catastrophic forgetting" if all layers are trained at once. Since their parameter space is so small, forcing the model to learn a new dataset can wipe out the previous weights in previous trainings. Progressive Fine-Tuning acts as a structural safeguard. By only unlocking specific blocks of the network at a time, it restricts the amount of the network that can change at once. This prevents the lightweight model from losing its generalized edge-detection skills while it learns the new cassava leaf data Chen and Liu (2025).

2.2 Review of Related Studies

2.2.1 MobileNetV3 Small

Zhang et al. (2025) developed an enhanced MobileNetV3 Small model for potato leaf disease diagnosis. The researchers incorporated a lightweight multi scale fusion

module. The network achieved a 94.89 percent accuracy on leaf datasets. The system provides a cost effective solution for resource limited field environments.

Jiao et al. (2025) created a lightweight network inspired by MobileNetV3 Small. The network integrated inverted residual blocks and depthwise separable convolutions to enhance feature extraction. The model provided a practical solution for precision agriculture. The researchers confirmed predictions focused on biologically relevant lesion regions. Bahaa et al. (2026) introduced a smart mobile application for plant disease detection. The system uses a lightweight architecture to focus on leaf regions. The model easily detects plant diseases through visual symptoms. The evaluation showed the system detects infections with high accuracy.

Roseno et al. (2024) compared ResNet50, VGG16, and MobileNetV3 Small for rice disease detection. The researchers trained the models on leaf datasets to evaluate classification accuracy. MobileNetV3 Small provided the highest test accuracy. The study concluded MobileNetV3 Small operates efficiently as the superior model for classification tasks. Not only this, but also this architecture ensures fast processing for field applications.

Zhou et al. (2025) conducted a comparative analysis of lightweight models for rice disease identification. MobileNetV3 Small showed competitive performance against heavier architectures. The smaller model attained a peak accuracy exceeding 99 percent.

The results underscore the potential of lightweight architectures for rapid mobile diagnostics.

Mvula and Masabo (2024) tested a pre trained MobileNetV3 model for in field tomato disease detection. The model achieved a 92.59 percent accuracy on controlled datasets. Testing on a primary field dataset resulted in a significant performance drop to a 9.2 percent accuracy. The study proved models trained on controlled datasets fail to generalize in real world scenarios.

2.2.2 Utilization of CLAHE and D CLAHE

Scientists evaluated a hybrid machine learning technique for cassava disease recognition. The researchers noted that images captured in natural environments suffer from poor illumination and low contrast. The researchers applied Contrast Limited Adaptive Histogram Equalization to enhance the images before segmentation. The system achieved high classification accuracy of 97% on its testing thus proving contrast enhancement resolves illumination issues in agricultural datasets (Nisha & Vijayakumar, 2022).

Tribiakto et al. (2024) developed an enhanced classification model for cassava leaf diseases. The research team identified a specific gap regarding practical deployment in agricultural diagnostics. Existing models struggle with uneven image quality and imbalanced class distributions. Poor image quality impairs feature extraction. The researchers integrated Contrast Limited Adaptive Histogram Equalization with

resampling techniques to address these limitations. The enhanced EfficientNetV2B2 model achieved an 88.1 percent classification accuracy. This confirms contrast enhancement and progressive training resolve variations in field images.

Ojo and Zahid (2023) investigated performance improvements resulting from various preprocessing and class imbalance strategies. The researchers evaluated techniques including adaptive histogram equalization and image sharpening. The evaluation findings indicate Contrast Limited Adaptive Histogram Equalization combined with image sharpening performed best. The system achieved an average classification accuracy of 97.69 percent. The study proves this specific combination reduces noise while enhancing structural edges.

To address the limitations of static image enhancement, Campos et al. (2019) utilized machine learning to dynamically determine optimal CLAHE hyperparameters, proving that the use of adaptive, image-specific parameters significantly improves overall model performance compared to manual tuning.

Liu et al. (2022) introduced the integration of a CNN-PP (Convolutional Neural Network Parameter Predictor) module. This parameter predictor analyzes global image contents to continuously tune preprocessing operations, ensuring optimized enhancement for each individual input.

Otsuka et al. (2026), The researchers introduced an Image-Adaptive CLAHE (IA-CLAHE) framework that mathematically establishes the differentiability of the

histogram redistribution process. The resulting system dynamically estimates optimal tile-wise clip limits directly from the input image. Because the operation is differentiable, it allows the enhancement mechanism to be trained end-to-end alongside the primary network.

2.2.3 Utilization of Layer Wise Progressive Fine Tuning

Wang et al. (2026) applied a stage wise progressive fine tuning strategy to large models for soybean leaf disease recognition. The research team introduced the dataset in stages. The model captured core task features before adapting to complex distributions. The progressive fine tuning improved the model accuracy to 93.82 percent. This proves progressive training stabilizes learning under small sample conditions.

Li and Sun (2023) proposed a layer wise fine tuning method to identify base and novel classes. The researchers applied this technique to a vision language model. The method improved classification accuracy by 5.01 percent over baseline models. The system recognizes image classes effectively in few shot scenarios.

Jiao et al. (2025) presented an interpretable approach for strawberry leaf disease recognition using a lightweight architecture. The researchers applied selective layer wise fine tuning to optimize the model for mobile deployment. The model achieved a 96.6 percent accuracy rate. This challenges the assumption requiring massive parameter updates for high accuracy.

Dong et al. (2024) investigated the impact of fine tuning paradigms on unknown plant disease recognition. The researchers compared visual prompt tuning against fully fine tuning and linear probe tuning. Visual prompt tuning outperformed the other methods in few shot scenarios. The model achieved a 94.8 percent accuracy rate. This proves appropriate fine tuning paradigms improve out of distribution detection performance directly.

Cig et al. (2023) proposed an adaptive contrast enhancement model for binary disease classification. The researchers compared standard equalization methods against a modified Contrast Limited Adaptive Histogram Equalization algorithm prior to network fine tuning. The enhanced images enabled the model to achieve a 96.16 percent accuracy. The modified technique outperformed baseline methods by optimizing image quality.

2.3 Concept of the Study

Figure 4

Conceptual Framework

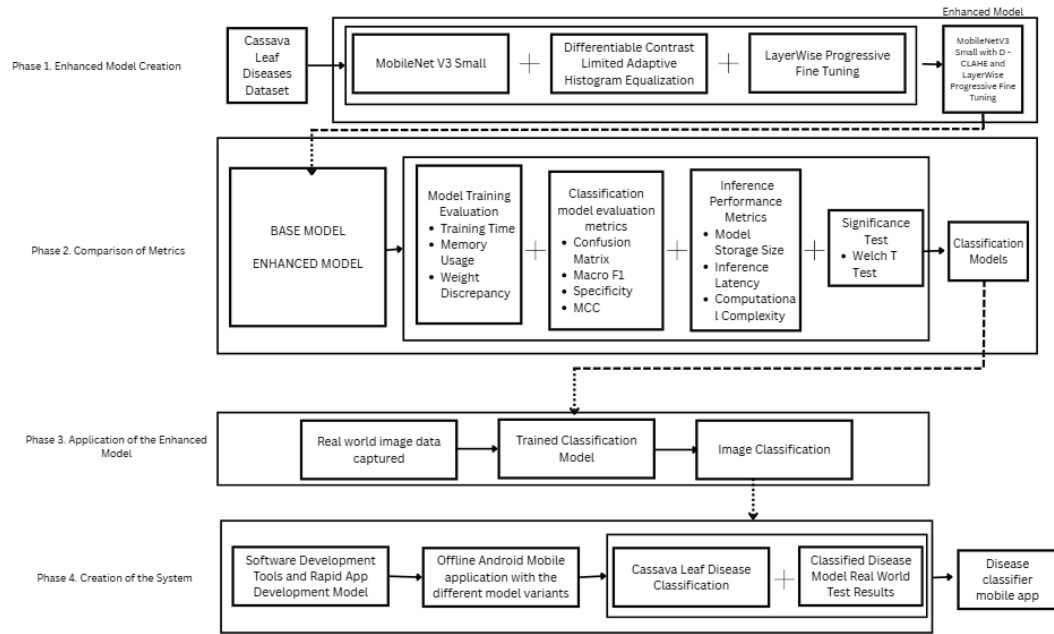


Figure 4 shows the conceptual framework that will be used on the study, its different phases will be divided into 4, The enhanced model creation, comparison of metrics, application of the enhanced model, then the creation of the system.

1. Phase 1 indicates the proposed addition to the base MobileNetV3 small's architecture and training process with D-CLAHE and Layerwise Progressive Fine Tuning during training of the model using the input dataset.
2. Phase 2 shows the multiple evaluation metrics to be used on the different model types to check if there's a difference in the respective measures, this is paired with the significance test to check the enhanced model impact significance.
3. Phase 3 shows the application of the enhanced model to its proposed application in the real world tests for image classification.

4. Phase 4 shows the development of the application for testing and validation of the enhanced model to its proposed environment and a chosen testing in classification of different cassava diseases to create a field ready mobile application

2.4 Synthesis

Studies such as those by Ren et al. (2025) and Abiyyu & Rahardi (2026) validate the efficiency of the MobileNetV3 architecture for edge devices. However, their methodologies rely on standard transfer learning and basic image normalization, making the models highly vulnerable to the issues and problems when exposed to harsh, real world lighting or localized disease strains. In addition, researchers like Hoang et al. (2024) utilized CLAHE to neutralize environmental lighting, and Patel et al. (2025) utilized Progressive Fine-Tuning to learn complex morphological diseases. Yet, both of these studies relied on massive, computationally expensive heavyweight CNNs like RESNET50 deployed on servers or cloud-dependent applications, making them inaccessible to farmers operating in remote, offline areas with mobile devices with minimal computational capabilities.

To bridge these gaps, recent advancements point to the necessity of dynamic parameter estimation (Campos et al., 2019) combined with efficient CNN-PP parameter prediction (Liu et al., 2022). With this, the proposed study fills this specific technological gap by integrating a novel, fully differentiable D-CLAHE layer directly into the MobileNetV3-Small architecture. By embedding D-CLAHE into both the forward and backward propagation loops, the model dynamically learns localized contrast equalization, eliminating the need for flawed manual tuning and parameter guessing. This

dynamic brightness suppression is paired with the structural weight-stabilization of Layer-Wise Progressive Fine-Tuning, ensuring consistent classification accuracy on low-power edge devices despite fluctuating environmental noise.

2.5 Definition of Terms

1. Artificial Intelligence (AI). Artificial Intelligence refers to the system to be developed using deep learning to execute real time disease classification without any human intervention.
2. Backpropagation. The specific computational process executed during the model's training phase where the system calculates the loss between its predicted outputs and the actual labels, and subsequently updates the weights of the MobileNetV3-Small and D-CLAHE modules to minimize this error.
3. Catastrophic Forgetting. Measured in this study as a significant drop in the model's validation accuracy on baseline or generalized features immediately following its training on the new, highly specific cassava leaf disease dataset.
4. Convolutional Neural Network (CNN). The deep learning architecture to be used in this research to extract spatial features (edges, textures, shapes) from the cassava leaf diseases image dataset for the purpose of classification.
5. CNN Parameter Predictor (CNN-PP). A secondary sub-network to be implemented inside the MobileNetV3- Small that dynamically outputs and adjusts the tuning parameters for the image enhancement module during the forward pass of each training step.

6. Differentiable Contrast Limited Histogram Equalization (D-CLAHE). A custom, trainable layer integrated directly into the neural network pipeline that automatically adjusts the localized contrast of input images on its histogram bins and clip limits.
7. Fine Tuning. A specific transfer learning technique where the existing mathematical weights of MobileNetv3 Small are further trained and adjusted on a new domain-specific dataset.
8. Gradient Shock. A training issue in neural networks where a sudden, excessively large gradient update destabilizes the model. This abrupt shift can disrupt convergence, destroy previously learned weight representations, or cause a severe spike in training loss and can be mitigated by the proposed application Layer Wise Progressive Fine Tuning during model training.
9. Histogram Equalization. A global image enhancement technique proposed to be applied to equalize the pixel brightness and contrast intensity distribution of the input images.
10. Image Classification. The algorithmic process by which the MobileNetV3-Small analyzes an image and classifies it among pre-determined classes.
11. Layerwise Progressive Fine-Tuning. The specific training protocol to be applied in this study where the model's pre-trained weights are updated in sequential phases starting by freezing the base layers to train only the classification head, and systematically unfreezing deeper blocks.
12. MobileNetV3-Small. The lightweight neural network architecture to be used as the primary classifier in this study and the preprocessing layer to be enhanced,

selected and evaluated based on its computational efficiency, inference speed, and suitability for deployment in mobile application.

13. Overfitting. To be identified during the training phase when the model's training accuracy continues to rise while its validation accuracy plateaus or decreases over 10 consecutive epochs, which triggers the early-stopping and reverting to the best epoch.
14. Parameters. The numerical weights and biases within the network's layers that are to be optimized during training and to be automatically adjusted in the D CLAHE preprocessing by the CNN PP.
15. Transfer Learning. The method of initializing the MobileNetV3-Small model with weights pre-trained on a large-scale dataset by ImageNet rather than randomized weights, before officially beginning the training process on the project's specialized dataset.

CHAPTER 3

OPERATIONAL FRAMEWORK

3.1 Materials

3.1.1 Software

System Architecture

Figure 5

System Architecture

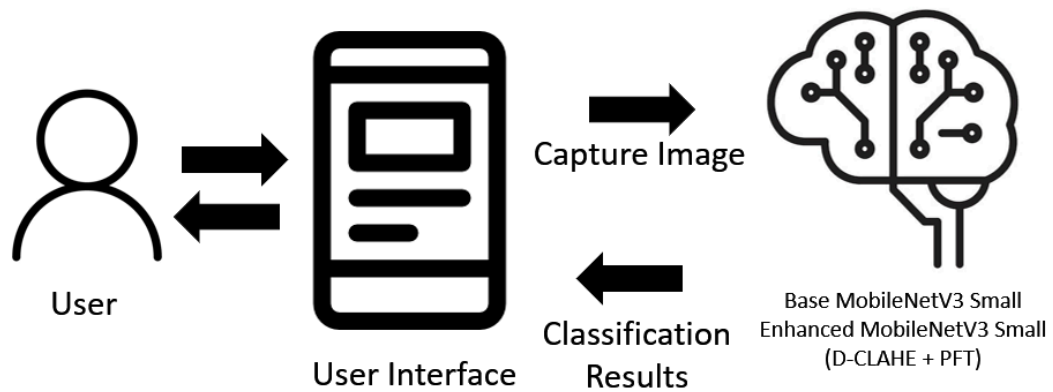


Figure 5 illustrates the system architecture of the offline mobile application, engineered specifically as a technical Proof of Concept (PoC) to test the models in a real-world edge environment. The architecture is designed to host and evaluate all the base MobileNetV3 Small and Enhanced MobileNetV3 Small simultaneously. When the user captures an image, the image will be sent to the models to be analyzed and classified. The classification results will be displayed on the user interface which the user will be able to view.

Table 1

Software Specification

Software Specification	Propose
Operating System	Development: Windows 10/11 to develop the enhanced MobileNetV3-Small Deployment: Android 10.0+ for the deployment of the mobile app for testing of the models.
Programming Language	Python: to be used for the Enhancement of MobileNetV3 Small Kotlin: to develop the android mobile app for testing
Libraries	Model Training and Enhancement: TensorFlow and Keras Libraries for MobileNetV3 Small enhancement and model training Model Evaluation: Numpy, Matplotlib, Seaborn, Sklearn
Integrated Development Environment (IDE)	Model Enhancement: Pycharm Mobile App Development: Android Studio Model Training: Google Colab

Details of the technology to be used:

Windows 10/11 will serve as the primary host environment for local script development and Android compilation. Android 10.0 or higher is designated as the target deployment environment, ensuring the final application is compatible with the modern mobile architectures. Python will be used as the primary language for deep learning and

mathematical data manipulation, while Kotlin will be used for developing the Android mobile app to be used for testing the models. TensorFlow and Keras libraries will be used to inject the custom D-CLAHE layer, RGB to YUV conversion and manage the 3-stage Layer-Wise Progressive Fine-Tuning mathematics without triggering gradient shock. The Numpy, Matplotlib, Seaborn and Scikit-Learn libraries will execute the quantitative analysis of the ablation study. NumPy will handle multi-dimensional array processing; Matplotlib and Seaborn will plot the training loss and validation accuracy curves; and Scikit-Learn will generate the confusion matrices and calculate the precision, recall, and F1-score metrics. PyCharm will be utilized for locally drafting and testing the enhanced model. Android Studio will serve as the primary IDE for developing the mobile application for testing the base and enhanced models. Due to the heavy computational overhead, Google Colab will be utilized to access the cloud-based T4 GPU acceleration required to train the models efficiently without hardware bottlenecking.

3.1.2 Hardware

Table 2

Hardware Specification

Device Specification	Propose
Processor: Ryzen 5	Training Setup: Google Colab Cloud CPU paired with a dedicated NVIDIA T4 Tensor Core GPU
RAM: 16GB	Training Setup: 12GB+ System RAM and 16GB GPU VRAM (To prevent dynamic tensor generation crashes)

Storage: 500GB SSD	Training Setup: Minimum 5GB Google Drive Cloud Storage (For raw datasets, augmented batches, and epoch checkpoints)
Mobile Device: Standard Consumer Smartphone	Deployment Setup (Minimum): 4-core CPU, 4GB+ RAM, 100MB Free storage, 4MP Camera

Details of the technology to be used:

Standard CPUs cannot handle the millions of Multiply-Accumulate Operations (MACs) required for deep learning. The Google Colab's T4 GPU is specifically required to handle the severe computational load of the model training. To handle the data and accommodate the training of the models and prevent crashes during training with ample overhead 12 GB System RAM and 16GB GPU VRAM is recommended. To accommodate the dataset and the codebases to be used in training in google colab at least 5GB free space in Google Drive is recommended. The deployment mobile phone hardware is recommended to have 4GB+ of RAM and at least 100MB of free storage to accommodate the app.

3.1.3 Data

The data utilized in this study will encompass cassava leaves classified into five specific target states: healthy asymptomatic leaves, Cassava Mosaic Disease , Cassava Bacterial Blight , Cassava Green Mottle, and Cassava Brown Streak Disease

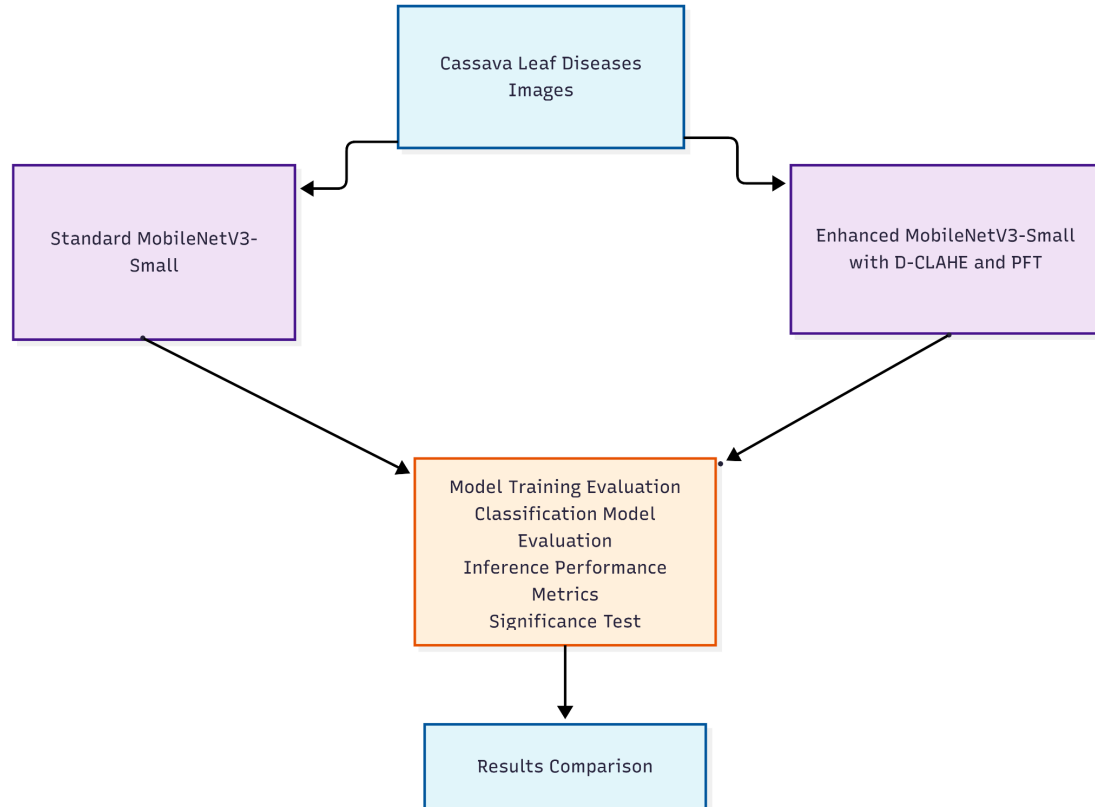
Data will be sourced from the verified Kaggle Cassava Leaf Disease Classification public dataset. This repository supplies verified images specifically to establish baseline weights. The data will also be supported by manually collected images across cassava farms in the municipalities of Echague and Ilagan using standard consumer grade

smartphone cameras to mimic the real world conditions of uncontrolled environmental noise such as harsh sun glare, contrast, and shadows. Capturing these are important to be able to stress test the D-CLAHE's ability to normalize contrast and improve model performance. The data will be resized to 224x224 to match MobileNetV3 Small's input and will undergo spatial augmentation which includes random flipping, zooming and rotating, the data will also be split into 80% for training, 10% for validation and 10% for testing Zhang et al. (2025).

3.2 Methods

3.2.1 Experimental Design

The research design/conceptual framework to be used in this study will involve 4 phases that aims to solve the proposed objectives using a Quantitative Experimental design. The phases focuses on enhancing the capabilities of the MobileNet V3 small by integrating a custom D-CLAHE layer in MobileNetV3 small and utilization of LayerWise Progressive Fine Tuning during training to fix the issue of the real world drop in performance of the model.

Figure 6**Experimental Design**

As shown in Figure 6, the study will compare the base MobilenetV3 Small and the enhanced MobilenetV3 with D-CLAHE and Progressive Fine Tuning. The models will be trained on cassava leaf disease images and will have three metric types for evaluation which are Model Training Evaluation, Classification Evaluation and Inference Performance.

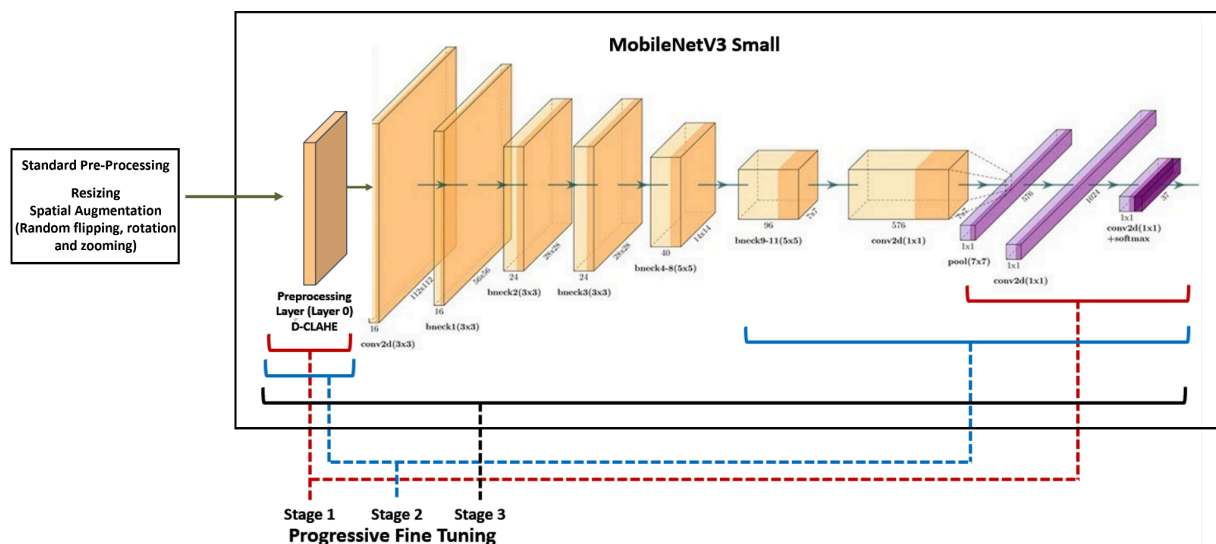
3.2.2 Procedures for the different Phases

3.2.2.1 Phase 1. Enhanced Model Creation

This initial phase focuses on the architectural modification and optimization of the core classification engine. The standard MobileNetV3-Small architecture will be utilized as the foundational feature extractor. To address the model's vulnerability to environmental lighting, a custom Differentiable CLAHE block is integrated natively into the model to perform dynamic contrast enhancement and edge sharpening during the forward pass. Following the architectural modification, the model undergoes Layer-Wise Progressive Fine-Tuning. This training approach unfreezes the network layers in 3 stages: stage 1 where only the custom outer layers are trained and the other layers are frozen, stage 2 where the deep bottleneck layers are unfrozen and stage 3 where all layers are unfrozen for a final global micro-adjustment of the entire architecture allowing the model to learn the specific features of cassava diseases without inducing catastrophic forgetting of its original pre-trained ImageNet weights. These structural and training modification results in the Enhanced Model, figure 7 shows the details of the whole process.

Figure 7

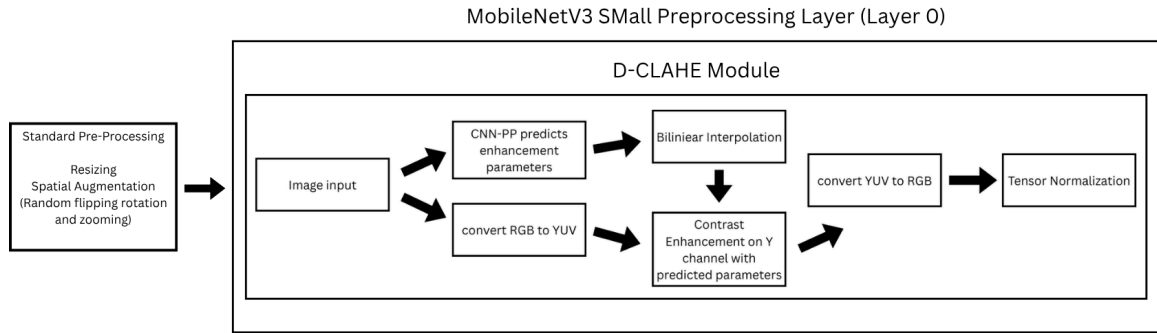
Proposed Enhanced MobileNetv3 Small Architecture



Trainable Frontend Enhancement

Figure 8

Trainable Differentiable-CLAHE Frontend Module



Layer 0 represents the newly integrated frontend of the modified MobileNetV3-Small architecture. Unlike traditional static preprocessing, this layer utilizes a Differentiable Contrast Limited Adaptive Histogram Equalization (D-CLAHE) module. Because it is embedded directly into the neural network's computation graph, it acts as an active, trainable feature enhancer prior to standard tensor normalization.

To manipulate contrast without distorting the biological color signatures of the cassava lesions, the RGB input tensor is explicitly converted into the YUV color space. This isolates the brightness of the image (Luma) from its colors (Chrominance).

$$Y = (0.299 \times R) + (0.587 \times G) + (0.114 \times B)$$

$$U = -(0.147 \times R) - (0.289 \times G) + (0.436 \times B)$$

$$V = (0.615 \times R) - (0.515 \times G) - (0.100 \times B)$$

Equation 1. RGB to YUV Color Space Conversion

By extracting the Y (Luma) channel, the module can dynamically correct harsh environmental shadows and sun glare common in agricultural environments, while the U and V channels remain untouched.

Traditional CLAHE relies on creating discrete histogram bins and computing Cumulative Distribution Functions (CDFs) to limit contrast. However, these sorting and counting operations are non-differentiable, which breaks the backpropagation chain required for neural network training (Otsuka et al., 2026) . To solve this, the proposed architecture replaces discrete histograms with a fully differentiable algebraic mechanism driven by a localized Convolutional Neural Network Parameter Predictor (CNN-PP).

The CNN-PP analyzes the normalized input image and outputs continuous, learnable parameters specifically a contrast modifier (α) and a brightness modifier (β) mapped across an 8×8 spatial grid. This results in 64 distinct enhancement tiles. Simultaneously, the isolated Y channel undergoes localized mean shifting via average pooling to establish a local baseline (μ). The contrast enhancement is then algebraically applied to the Y channel as follows:

$$Y_{enhanced} = (Y - \mu) \times (1.0 + \alpha \times 0.75) + \mu + (\beta \times 0.25)$$

Equation 2. Differentiable Algebraic Contrast Enhancement

In this equation, the network algebraically limits the contrast to prevent noise amplification, while allowing the system to learn the optimal lighting conditions for disease identification during backpropagation. The enhanced pixels are subsequently clipped between $[0.0, 1.0]$ to prevent numerical overflow.

To prevent grid artifacting or harsh boundaries between the 64 tiles, bilinear interpolation is applied to the predicted parameters, ensuring smooth transitions across

the leaf surface. Finally, the dynamically enhanced Y channel is concatenated back with the preserved U and V channels. The tensor is mathematically reconstructed into the standard RGB color space, where it seamlessly enters the first convolutional filter of the MobileNetV3 backbone for feature extraction.

Convolutional Neural Network Parameter Predictor (CNN-PP)

To power the algebraic contrast enhancement described in Equation 2, the D-CLAHE module relies on an auxiliary neural network known as the Convolutional Neural Network Parameter Predictor (CNN-PP) which adapts the concept demonstrated by Liu et al. (2021) . The CNN-PP acts as a dynamic spatial lighting analyzer. By assessing the unique photometric conditions of each specific input tensor, it allows the network to predict localized contrast (α) and brightness (β) modifiers. This mechanism enables the architecture to actively learn how to illuminate the leaf lesions to maximize the feature extraction capabilities of the downstream MobileNetV3 Small backbone, making it end-to-end differentiable. The CNN-PP is implemented as a lightweight, parallel feature extractor that processes the normalized RGB input tensor immediately upon entry. To ensure the model remains highly efficient for eventual edge-device deployment, the predictor utilizes a compact progression of downsampling convolutional layers rather than a heavy, deep network.

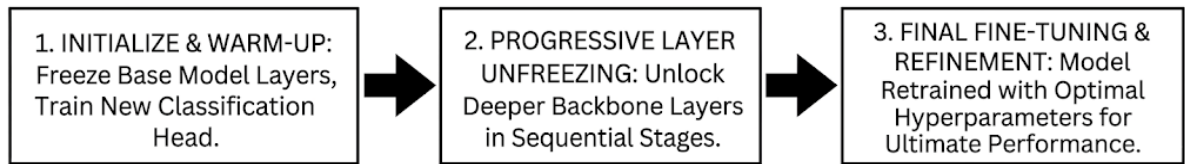
Layer Wise Progressive Fine Tuning

Dense Head (Feature Extraction)

On the dense layer and classification layer the proposed Layerwise Progressive Fine Tuning will be applied and will have the following steps:

Figure 9

Layer Wise Progressive Fine Tuning Step Diagram



The new Layer 0 and the created 5 dense head for each classes are still uninitialized, with its starting weights being randoms creating initial error thus the entire MobileNetV3-Small convolutional backbone is frozen and the gradient calculation for all backbone weights and is forcibly zeroed out during backpropagation as to not affect its performance. Only the weights of the newly added dense classification head are unfrozen. The unfrozen dense classification head is then trained on ADAM optimizer using the stage 1 high learning rate, aggressively learning the patterns for the first network to be stable until the gradients match to the MobileNetV3 Small as the new dense head error rate drops close to 0.

Classification Layer

As the new dense head stabilizes, the model needs to learn the specific and complex edges and geometries of the dataset. On the process of layer wise fine tuning, the other layers will still be frozen but the Bottleneck Layer will be unfrozen for it to

learn, and the learning rate will have the stage 2 rate lower than stage 1 to prevent gradient shock. Since the new learning rate is so low, the weights will shift slowly, this will ensure the pre-trained filters will see the cassava leaf lesions without forgetting the original ImageNet edge detection training.

In the final step as the crucial layers are trained, the entire network will be unfrozen with an even lower learning rate minimizing the loss function across the parameters needed for identification. At this stage the expected gradients will be small as the model will converge on the global minimum making the whole model to work as one on its layers and will be monitored using the validator for overfitting.

Optimizer and Loss Function

The optimization of the neural networks parameters on weights and biases is a crucial step for the optimal values that result in the lowest possible error or loss. This study will use a type of optimizer called Adam (Adaptive Moment Estimation) optimizer and it uses a combination of the concepts of momentum and adaptive learning rates. The optimizer improves training speeds in deep neural networks for the neurons to converge for learning. It customizes each parameter's learning rate based on its gradient history, and this adjustment helps the neural network learn efficiently as a whole Agarwal (2023). For the model to evaluate its own performance on probability distribution on all 5 target classes during training, it will be calculated using the Categorical Cross Entropy Loss Function seen in Equation 3.

$$L(y, \hat{y}) = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

Equation 3. Categorical Cross-Entropy loss function

This function where C is the number of classes where as y_i is the ground truth and the hatted y sub i is the predicted probability, since the logarithmic function apply a penalty to the network for predictions that are both highly confident and incorrect, this forces the MobileNetv3 small model to learn more the difference between visually similar diseases and learn to adjust the D-CLAHE parameters accordingly.

3.2.2.2 Phase 2. Comparison of Metrics

Phase 2 will have the evaluation of the four trained models and will assess their performance in multiple metrics during and after the training and will have the following evaluation metrics.

Model Training Evaluation

The validation set will have the preliminary check if there's an improvement to the process of applying D-CLAHE and Progressive Fine Tuning to the MobileNetV3 Small can have benefits compared to its base model.

1. Training Time - The training time will be checked and timed on the Google Collab with the constant hardware across all training sessions.
2. Memory Usage - The memory usage required for training will be checked and monitored on the Google Collab with the constant hardware across all training sessions.
3. Weight Discrepancy/L2 Norm - To prove that Progressive Fine-Tuning preserved the foundational ImageNet edge detectors better than standard fine-tuning, the L2 norm of the weight vectors in the foundational layers is calculated before and after training using the Equation 4. A lower L2 distance between the start of

training to after training is better and can validate if the proposed process mitigates gradient shock

$$D(W_{base}, W_{ft}) = ||W_{base} - W_{ft}||_2$$

Equation 4. Weight Discrepancy Distance

Classification Models Evaluation Metrics

The primary analytical tool utilized is the Confusion Matrix that consists of the Accuracy Precision Recall and F1 Score, from which the core diagnostic metrics are used for all model variants to be used for comparison. This will also be used on the mobile application testing of the models in the field.

$$Accuracy = \frac{True\ Positive}{TP + TN + FP + FN}$$

Equation 5. Accuracy

$$Precision = \frac{TP}{TP + FP}$$

Equation 6. Precision

$$Recall = \frac{TP}{TP + FN}$$

Equation 7. Recall

$$F1\ SCORE = 2 * \frac{PRECISION * RECALL}{PRECISION + RECALL}$$

Equation 8. F1 Score

1. The accuracy shown in equation 5 will be used to evaluate the overall correctness of the classification of each class. Equation 6 shows the precision evaluation of

the model and will be used to ensure that there are minimal false positives. Recall as shown in Equation 7 will be used for the sensitivity of the model, this an important metric in disease classification since missing a positive disease and classifying the leaf as a normal leaf can have a huge consequence. And lastly, the F1 Score will be used to check if the accuracy is misleading checking for imbalances and verifying the accuracy results (*Mastering the F1 Score: A Practical Guide for Machine Learning Success*, n.d.) (*Classification: Accuracy, Recall, Precision, and Related Metrics*, n.d.)Jayaswal (2025).

2. Macro Averaged F1 Score - This will be used to determine the mean of the Precision and Recall for each specific diseases then averages them, a high score can mean good performance of the classes:

$$Macro\ F1 = \frac{F1_1 + F1_2 + F1_3 + \dots + F1_N}{N}$$

Equation 9. Macro F1 score

3. True Negative Rate: This evaluates the model's ability to correctly identify a completely healthy cassava leaf. A high specificity rating mathematically proves that the model is highly resistant to triggering false alarms which is critical to preventing farmers from unnecessarily applying chemical treatments to healthy crops.

$$Specificity = \frac{TN}{TN+FP}$$

Equation 10. True Negative Rate

4. Matthews Correlation Coefficient: To validate the probabilities of the model, the MCC will be calculated. Unlike standard metrics, MCC evaluates all four quadrants of the confusion matrix simultaneously, producing a value between -1.0 and +1.0. A high MCC score will guarantee that the model have true predictive across both the majority and minority classes, rather than simply guessing the most common class.

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

Equation 11. MCC Score

3.2.2.3 Phase 3. Application of the Enhanced Model

The phase 3 of the study focuses on the application of the enhanced model with D-CLAHE and LayerWise Progressive Fine Tuning applied to a real world test using actual field captured images and testings in actual mobile application using 4 different kinds of cassava leaf diseases in Isabela Province determining the performance of the model to its intended uses with the optimization in harsh lighting scenarios typically found in real world tests.

3.2.2.4 Phase 4. Creation of the System

As to test the application of the enhanced model, an android mobile app will be created as a testing tool and comparison basis for the different models trained and a released version will be created for the end users with the intended functions and interface.

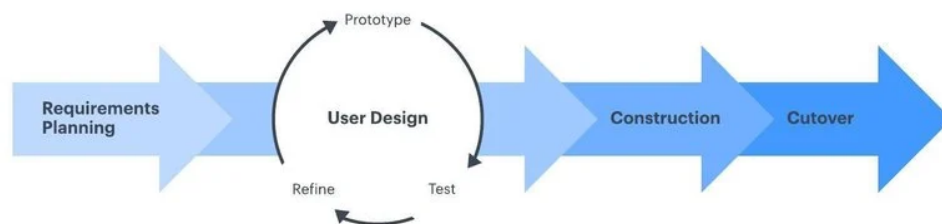
The mobile app will use the models mentioned to be tested to classify 4 kinds of cassava leaf diseases and healthy leaves to automate disease classification in the cassava fields in Isabela Province.

System Development

The system will be developed through the Rapid Application Development Model to ensure fast deployment and testings of the different classification models to find the most suitable for the needs of the users (Rouse, 2016).

Figure 10

Rapid Application Development Process



Utilizing Software Development Tools and following the Rapid Application Development (RAD) model, an offline Android mobile application is constructed. This application is engineered to embed the different model variants, acting as a technical Proof of Concept (PoC) testing environment. The system allows agricultural users to capture cassava leaf images via the smartphone camera and run the classification models entirely offline. Finally, the application processes the tensor data, executes the Cassava Leaf Disease Classification, and outputs the Classified Disease Model Real World Test

Results directly to the user interface, proving the architecture's viability for remote agricultural edge computing.

3.2.3 Evaluation

Inference Latency and Memory Usage

1. On the usage of the mobile application in the real world testing the study will validate the system's performance for real-time edge computing on an offline Android application using inference latency and memory usage.
2. The models storage size will also be recorded to compare if there's a significant storage size increase or decrease to improve the performance of the MobileNetV3 Small and if the performance improvement amount can be justified on the storage size increase.
3. Inference latency is a metric that will be used to check the delay between image capture, and the results of the classification. This will be measured in milliseconds (ms) and tracks the time complexity of the models to be compared.
4. Computational Complexity will be used as while inference latency fluctuates depending on the specific model of the smartphone being used, Multiply-Accumulate Operations (MACs) mathematically quantify the computational work the CPU is doing. Measuring the MACs proves that the utilization of Depthwise Separable Convolutions fundamentally lowers the mathematical cost of the network, making it hardware-agnostic.

$$COST_{depthwise} = (DK * DK * M * DF) + (M * N * DF * DF)$$

Equation 12. Multiply-Accumulate Operations

Significance Test

1. Welch T Test - To ensure the performance of the enhanced model are not the result of random variance during weight initialization, Welch's t-test is applied to the accuracy distributions of multiple training runs and a p value of less than 0.05 will be needed to be observed to observe significance

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

Equation 13. Welch T test

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